

Community Mod Framework

Community Mod Framework

Community Mod Menu

Community Mod Toolkit

Game modification: Community Mod Framework

İçindekiler

Overview

Basic Setup

Features

- [Community Mod Menu \(CMM\)](#)
- [Action Bar](#)
- [Custom Alerts](#)
- [Mod Banners](#)
- [Mod Action Log](#)
- [On-Action Hooks](#)
 - [Game Start & Load Hooks](#)
 - [Country Transfer](#)
 - [Recurring Hooks](#)
- [is_host Trigger](#)
- [cmf_can_use_unrestricted_tools Trigger](#)
- [Mod Detection](#)
- [cmf_suppress Effect](#)
- [cmf_change_variable_map Effect \(Deprecated\)](#)
- [GUI Macros](#)
- [Improved Vanilla Top-Level Widget Overrides](#)
- [Community Mod Toolkit \(CMT\)](#)

Additional Configuration

- [Conditional Visibility](#)
- [Dependency Check Popup](#)

Links

Community Mod Framework



Type
Mod Framework & Toolkit

Author(s)
Bahmut, CaesarVincens, Conner, Glorp, Pickle, RomanImperator

Status
Released



The Community Mod Framework (CMF) is a shared mod framework for Europa Universalis 5 that provides a set of features mods can use to integrate with the game without conflicting with each other. It includes a shared in-game mod menu, a custom action bar, dynamic alerts, utility triggers, and a companion development toolkit with a visual editor, workshop uploader, and mod translator.

The framework is designed to be invisible when no dependent mods are active.

CMF is available on the [Steam Workshop \(https://steamcommunity.com/sharedfiles/filedetails/?id=3692202776\)](https://steamcommunity.com/sharedfiles/filedetails/?id=3692202776). For full documentation, see the [GitHub wiki \(https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/wiki\)](https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/wiki).

Basic Setup

Hook your mod into CMF's shared registration on_action. This runs for each human player on game start and when the mod menu opens. Not all features require it, but it is the standard entry point for registering CMM settings, action bar buttons, and alerts.

```
# in_game/common/scripted_effects/your_effects.txt
your_mod_register = {
    # Register settings, action bar elements, alerts, etc.
}

# in_game/common/on_action/your_on_actions.txt
cmf_on_mod_registration = {
    on_actions = {
        your_mod_on_register
    }
}

your_mod_on_register = {
    effect = {
        your_mod_register = yes
        # You can optionally split registration into as many effects as you like:
        # your_mod_register_cmm_settings = yes
        # your_mod_register_action_bar = yes
    }
}
```

Features

Community Mod Menu (CMM)

The Community Mod Menu is a shared in-game settings UI that allows multiple mods to hook into a single menu through a registration API. It supports toggles, sliders, dropdowns, buttons, text inputs, and settings lists.

See the [CMM page](#) for a quick reference.





work/wiki/CMM-Visual-Editor), which provides a graphical interface and a live preview to create your mod menu.

Action Bar

The Action Bar is a shared UI bar where mods can place interactive buttons. Players can reposition the bar (top, bottom-left, or bottom-right) and toggle individual buttons on or off from the Action Bar tab of the Community Mod Framework entry in the mod menu.

Each button requires localization keys:

```
l_english:
cmf_action_bar_element_example:
"cmf_action_bar_element_example"
cmf_action_bar_element_example_color: "gold"
cmf_action_bar_element_example_icon: "@advance!"
cmf_action_bar_element_example_name: "Some Button"
cmf_action_bar_element_example_tooltip: "This is a custom Action Bar Button."
```

Effect	Description
cmf_add_action_bar_element = { element = <root_key> }	Register a button
cmf_remove_action_bar_element = { element = <root_key> }	Remove a button

Handle clicks via cmf_on_callback on_action (var:cmf_callback = the element flag).

Hide the entire action bar from GUI code with

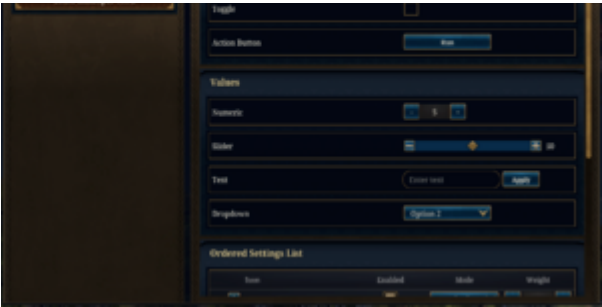
GetVariableSystem.Set('cmf_hide_action_bar', 'yes') and restore it with GetVariableSystem.Clear('cmf_hide_action_bar').

Custom Alerts

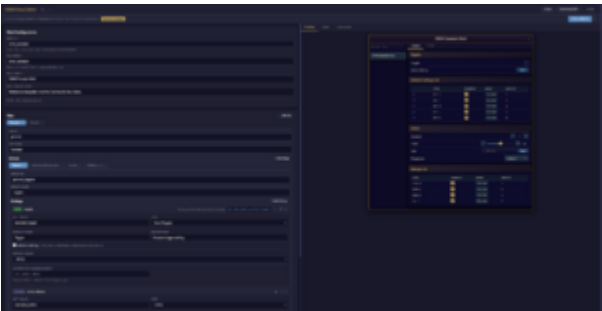
The Custom Alerts system allows mods to display dynamic, dismissable notifications in the in-game alert bar. Alerts are color-coded and interactive: players can click them to trigger an effect, or right-click to dismiss.

Each alert requires localization keys:

```
l_english:
cmf_alert_example_color: "orange"
cmf_alert_example_icon: "@advance!"
cmf_alert_example_name: "Some Alert"
```



The Community Mod Menu.



The CMM Visual Editor.



Custom action bar buttons.



Custom alerts in the alert bar.



<root_key>_color	blue, orange, red, red_war, black, yellow, green, or purple
<root_key>_icon	Text icon (e.g. @advance!)
<root_key>_name	Tooltip header text
<root_key>_tooltip	Tooltip body text

All effects require **country scope**.

Effect	Description
cmf_show_alert = { alert = <root_key> }	Show an alert
cmf_remove_alert = { alert = <root_key> }	Remove an alert
cmf_is_alert_active = { alert = <root_key> }	Check if an alert is active (trigger)

Handle clicks via `cmf_on_callback on_action` (`var:cmf_callback` = the alert flag). Left-click automatically removes the alert.

Mod Banners

Display your mod as a banner on the pre-game lobby screen, with an icon, name, and description. Opt in by calling `cmf_register_lobby_banner` from a leaf `on_action` under `cmf_on_banner_registration`.

```
cmf_on_banner_registration = {
  on_actions = {
    your_mod_register_lobby_banner
  }
}

your_mod_register_lobby_banner = {
  effect = {
    cmf_register_lobby_banner = { mod_id = your_mod }
  }
}
```

The banner icon and background are game concepts (`<mod_id>_banner_logo`, `<mod_id>_banner_background`); the title and description come from the `<mod_id>_name` and `<mod_id>_desc` localization keys.

Mod Action Log

The Mod Action Log provides a shared, global in-game log where mods can record actions. Each entry stores the actor country and localized action text, with optional named arguments for richer messages.

```
# Simple entry - actor is the current country scope
cmf_log = { action = my_mod_action_key }

# Entry with arguments - renders as: [arg1] [action] [arg2]
cmf_log_with_args = { action = transferred_to arg1 = paris arg2 = FRA }

# Clear all entries (disabled in multiplayer)
cmf_clear_log = yes
```



cmf_log_with_args = { action = <flag> arg1 = <flag> arg2 = <flag> }	Log with two named arguments (localization keys)
cmf_log_with_scope_arg = { action = <flag> }	Log with one country scope argument (requires scope:cmf_log_arg2)
cmf_log_with_scope_args = { action = <flag> }	Log with two country scope arguments (requires scope:cmf_log_arg1 and scope:cmf_log_arg2)
cmf_clear_log = yes	Clear all entries

All arguments are localization keys. The log is viewed via **Mod Menu > General > Session > Mod Action Log**.

On-Action Hooks

CMF provides shared on_action hooks. Hook into them the same way as cmf_on_mod_registration.

Game Start & Load Hooks

All fire once in no scope (like vanilla on_game_start). Each has a _human_country variant that fires in country scope for each human player.

Hook	Scope	When
on_game_start_after_lobby	none	Once on new game, after country selection
on_game_start_after_lobby_human_country	country	Same, per human country
on_game_load	none	Every save load (including from country selection)
on_game_load_human_country	country	Same, per human country
on_game_load_after_lobby	none	Every save load, only after passing country selection
on_game_load_after_lobby_human_country	country	Same, per human country

```
# No scope - for global effects
on_game_start_after_lobby = {
  on_actions = { your_mod_on_new_game }
}

# Country scope - for human country effects
on_game_start_after_lobby_human_country = {
  on_actions = { your_mod_on_new_game_country }
}
```

Country Transfer

cmf_on_country_transfer fires when the player switches countries (tag switches, inheritance, etc.). CMF auto-copies CMM settings. scope:old_country and scope:new_country are available.





Recurring Hooks

Slightly more efficient than each mod checking `is_ai = no` individually:

Hook	Frequency
<code>cmf_yearly_human_country_pulse</code>	Once per year
<code>cmf_monthly_human_country_pulse</code>	Once per month

```
cmf_yearly_human_country_pulse = {
  on_actions = {
    your_mod_on_yearly
  }
}

cmf_monthly_human_country_pulse = {
  on_actions = {
    your_mod_on_monthly
  }
}
```

is_host Trigger

CMF provides a trigger to check if the current country is controlled by the host player. Always evaluates to yes in singleplayer.

```
if = {
  limit = {
    is_host = yes
  }
  # Runs if the country is controlled by the host
}
```

cmf_can_use_unrestricted_tools Trigger

True when the "Enable Unrestricted Tools" setting (labelled "Enable Host-Only Tools" in multiplayer) is on and this country is the host (or in singleplayer). Use to gate restricted features not meant for normal play, e.g. unrestricted land transfer.

```
if = {
  limit = {
    cmf_can_use_unrestricted_tools = yes
  }
  # Runs if this country is the host and unrestricted tools are enabled
}
```





Mod Detection

cmf_is_mod_active checks whether another mod is active in the current game. Mods that register CMM settings are detected automatically; other mods call cmf_register_mod in their cmf_on_mod_registration hook.

```
if = {
    limit = { cmf_is_mod_active = { mod_id = some_other_mod } }
    # Runs only when some_other_mod is loaded
}

# Non-CMM mods: register so others can detect you
cmf_register_mod = { mod_id = your_mod }
```

cmf_suppress Effect

CMF provides a cmf_suppress effect to silence "used but never set" and "set but never used" engine warnings. Call inside an always = no dead branch:

```
if = {
    limit = { always = no }
    cmf_suppress = { v = my_variable }
    cmf_suppress = { v = my_flag_name }
}
```

Group multiple calls in one if block so the engine evaluates always = no only once.

cmf_change_variable_map Effect (Deprecated)

Deprecated because add_to_variable_map now overwrites an existing key directly.

Effect	Description
cmf_change_variable_map = { name = ... key = ... value = ... }	Update a country-scoped variable map entry
cmf_change_local_variable_map = { name = ... key = ... value = ... }	Update a local variable map entry
cmf_change_global_variable_map = { name = ... key = ... value = ... }	Update a global variable map entry

GUI Macros

```
visible = "[Nand(Foo, Bar)]" # not both
visible = "[Nor(Foo, Bar)]" # neither
visible = "[Xor(Foo, Bar)]" # exactly one
```

Improved Vanilla Top-Level Widget Overrides



widget itself. That reduces what you have to maintain and lowers the chance of conflicts with other mods that modify parts of the same file.

Covered files: `foreign_country_lateralview.gui`, `government_lateralview.gui`,
`ingame_topbar.gui`, `location_window.gui`, `outliner_entries.gui`,
`single_unit_window.gui`, `technology_lateralview.gui`.

Community Mod Toolkit (CMT)

The Community Mod Toolkit is a companion set of development tools for EU5 mod authors. It provides a structured mod template and automation tools that work together through shared configuration:

- **CMM Visual Editor** (<https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/wiki/CMM-Visual-Editor>) - Browser-based interface for designing CMM settings without scripting it by hand. The easiest way to get started with CMM.
- **Upload Tool** - Automated Steam Workshop deployment with submod support.
- **Translation Tool** - Automated localization via DeepL or Gemini AI for all EU5-supported languages.

See the [CMT](#) page for details.

Additional Configuration

Conditional Visibility

Any CMM setting or CMF element (action bar buttons) can optionally use a Scripted GUI for dynamic visibility or enabled state. Create a Scripted GUI named after the element and register it with `cmf_register_scripted_gui`:

```
# Register the GUI (call after adding the element):
cmf_register_scripted_gui = { element = cmf_action_bar_element_example }
```

```
# Action bar: is_shown controls visibility, is_valid controls enabled/greyed out
cmf_action_bar_element_example = {
    scope = country
    is_shown = { ... }
    is_valid = { ... }
}

# CMM setting: is_shown controls visibility (named [mod_id]__[setting_id]_on_changed)
your_mod__your_toggle_on_changed = {
    scope = country
    is_shown = {
        "variable_map(cmm|flag:your_mod__some_other_setting)" >= 1
    }
}
```

Dependency Check Popup



s) and extract into your mod folder. See the [GitHub wiki \(https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/wiki/Dependency-Check-Popup\)](https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/wiki/Dependency-Check-Popup) for details.

Links

- [Steam Workshop \(https://steamcommunity.com/sharedfiles/filedetails/?id=3692202776\)](https://steamcommunity.com/sharedfiles/filedetails/?id=3692202776)
- [GitHub Repository \(https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework\)](https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework)
- [Full Documentation \(GitHub Wiki\) \(https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/wiki\)](https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/wiki)
- [Example Mod \(GitHub\) \(https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/tree/main/submods/cmf-example-mod\)](https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework/tree/main/submods/cmf-example-mod)

Modding		Return to top
Documentation	Defines • Effects • Scopes • Scope links • Triggers Colors • Macros • Mean time to happen • Modifier types • On actions • Script value • Variables GUI script • Localization	
Scripted content	Actions • Disasters • Events • Missions • Modifiers • Scripted gui • Setup • Situations • Customizable localization	
Scripted types	Advances • Art • Buildings • Bureaucracies • Casus belli • Characters • Concepts • Countries • Culture • Diplomacy • Diseases • Estates • Goods • Institutions • International organizations • Laws • Movements • Peace treaties • Pops • Religion • Subject types • Traits • Units • Wargoals	
Map	Map • Map modes • Terrain	
Graphics	3D Models • Interface • Graphical assets • Fonts • Flags	
Audio	Music • Sound	
Other	AI • Console commands • Checksum • Mods • Mod compatibility • Mod structure • Troubleshooting	
Guides	Interface modding guide • Mod translation • Save-game editing • Settlement position modding guide	
Tools	Arcanum • PDX DeepL • PDX Workshop Manager • Community Mod Toolkit • Add Your Tool to the Wiki	

"https://eu5.paradoxwikis.com/index.php?title=Community_Mod_Framework&oldid=35034" sayfasından alınmıştır

Sayfa en son 17.31, 7 Temmuz 2026 tarihinde değiştirildi.

Aksi belirtilmedikçe içeriğin kullanımı Attribution-ShareAlike 3.0 lisansı kapsamında uygundur.





GAMES

[Discover](#)

[Our Brands](#)

[Browse](#)

[Play on Paradox technology](#)

ABOUT

[News](#)

[About us](#)

[Careers](#)

[Join our playtests](#)

[Media contact](#)

COMMUNITY

[Paradox Forums](#)

[Paradox Wikis](#)

[Support](#)

SOCIAL MEDIA

[Terms & Policies](#)

[Legal Information](#)

[EU Online Dispute Resolution](#)

[Report Illegal Content](#)

[Frequently Asked Questions](#)

[Paradox Interactive corporate website](#)

